Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 1 of 16 · find part c.iv in this mark scheme extract

1(a)	A1	**The only correct answer is D** ***A*** *is not correct because it is the definition of decomposition* ***B*** *is not correct because it is a task undertaken during evaluation* ***C*** *is not correct because it describes a method for expressing algorithms*		**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 2 of 16 · find part c.iv in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 3 of 16 · find part c.iv in this mark scheme extract

1b		Award **one** mark for each of:		
	B1	Initialisation of myName to "" (1)	• `String myName = "";` • `String myName = "";` • `myName = ""` Allow null Allow single space between quotes " "	
	B2	Add a line to create myAge and initialise it to 0 (1)	• `int myAge = 0;` • `int myAge = 0;` • `myAge = 0` Allow any method	
	B3	Complete the test by inserting < (1)	• `if (myAge < 30)` • `if (myAge < 30)` • `if (myAge < 30):` Allow <= Can award reverse test only if welcome message contained within an 'else' part	
	B4	Add a line to display "Welcome" (1)	• `Console.WriteLine ("Welcome");` • `System.out.println ("Welcome");` • `print ("Welcome")` ignore indentation Allow any method	**4**

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 4 of 16 · find part c.iv in this mark scheme extract

C#

```
// Q01bFINISHED

// Complete the line to initialise myName to an empty string
String myName = "";

// Add a line to create an integer variable called myAge and
//     initialise myAge to 0
int myAge = 0;

Console.WriteLine ("Enter your name: ");
myName = Console.ReadLine();
Console.WriteLine ("Enter your age: ");
myAge = Convert.ToInt32 (Console.ReadLine());

// Complete the test to check if the age is less than 30
if (myAge < 30)
{
    // Add a line to display a message that says "Welcome"
    Console.WriteLine ("Welcome");
}

```

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 5 of 16 · find part c.iv in this mark scheme extract

Java

```java
// Q01bFINISHED

import java.util.Scanner;
public class Q01bFINISHED
{
    public static void main (String[] args)
    {
        Scanner myKeyboard = new Scanner (System.in);

        // Complete the line to initialise myName to an empty string
        String myName = "";

        // Add a line to create an integer variable called myAge and
        //     initialise myAge to 0
        int myAge = 0;

        System.out.println ("Enter your name: ");
        myName = myKeyboard.nextLine();
        System.out.println ("Enter your age: ");
        myAge = Integer.parseInt (myKeyboard.nextLine());

        // Complete the test to check if the age is less than 30
        if (myAge < 30)
        {
            // Add a line to display a message that says "Welcome"
            System.out.println ("Welcome");
        }
    }
}
```

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 6 of 16 · find part c.iv in this mark scheme extract

Python

```
# Q01bFINISHED

# Complete the line to initialise myName to an empty string
myName = ""

# Add a line to create an integer variable called myAge and
#     initialise myAge to 0
myAge = 0

myName = input ("Enter your name: ")
myAge = int (input ("Enter your age: "))

# Complete the test to check if the age is less than 30
if (myAge < 30):

    # Add a line to display a message that says "Welcome"
    print ("Welcome")

```

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 7 of 16 · find part c.iv in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 8 of 16 · find part c.iv in this mark scheme extract

1c(i)		Award **one** mark for any of the following:		
	C1	C#: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Java: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Python: • # Q01c (1) • Q01c (1) • # Loop for 50 times (1) Loop for 50 times (1)	Ignore transcription errors	**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 9 of 16 · find part c.iv in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 10 of 16 · find part c.iv in this mark scheme extract

1c(ii)		Award **one** mark for any of the following:		
	C2	C#: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Java: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Python: • count = 0 (1) • total = 0 (1) • subtotal = 0 (1)	Ignore transcription errors	1
Question	**mp**	**Answer**	**Additional guidance**	**Mark**

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 11 of 16 · find part c.iv in this mark scheme extract

1c(iii)		Award **one** mark for:		
	C3	• if (1)	Applies to all programming languages Ignore capitalisation	**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 12 of 16 · find part c.iv in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 13 of 16 · find part c.iv in this mark scheme extract

1c(iv)		Award **one** mark for:		
	C4	C#: • && (1) Java: • && (1) Python: • and (1)	Ignore capitalisation	1

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 14 of 16 · find part c.iv in this mark scheme extract

1(d)	D1	**The only correct answer is B** ***A*** *is not correct because the data can be stored in an array as it is all the same data type (string)* ***C*** *is not correct because the data can be stored in an array as it is all the same data type (integer)* ***D*** *is not correct because the data can be stored in an array as it is all the same data type (real)*		**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 15 of 16 · find part c.iv in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 1 of 23 · Question 1 part c.iv · 1 mark

2024 JUN · Paper 02 · extract 16 of 16 · find part c.iv in this mark scheme extract

<table>
<tr>
<td>1(e)</td>
<td>E1
E2</td>
<td>One mark for each correct cell:
<table>
<tr><th>Item</th><th>Data type</th></tr>
<tr><td>45.82</td><td>All programming languages:
real/float
C#:
decimal/double
Java:
Double/double/BigDecimal
Python:
decimal</td></tr>
<tr><td>True</td><td>All programming languages:
Boolean/bool</td></tr>
</table></td>
<td>Ignore capitalisation
Do not accept string for True</td>
<td>2</td>
</tr>
</table>

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 1 of 13 · find part b.ii in this mark scheme extract

2(b)(i)	24.0 (1)	Accept 24	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 2 of 13 · find part b.ii in this mark scheme extract

2(b)(ii)	One of: By the rules of precedence multiplication is done before addition. (1) Without the brackets the addition would be done to the total of the multiplication. (1) By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1) newSpeed and intialSpeed have to be added together before multiplying (by time). (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 3 of 13 · find part b.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 4 of 13 · find part b.ii in this mark scheme extract

2(b)(iii)	Real (1) Because real x integer = real (1)	Accept double / float Accept 'because it could be a number with decimal places'	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 5 of 13 · find part b.ii in this mark scheme extract

2(b)(iv)	27.375 (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 6 of 13 · find part b.ii in this mark scheme extract

Q2(c)(i)	C#: 16 or 20 (1) Java: 16 or 21 (1) Python: 8 or 10 (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 7 of 13 · find part b.ii in this mark scheme extract

Q2(c)(ii)	C#: 29 or 31 (1) Java: 24 or 26 (1) Python: 14 or 15 (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 8 of 13 · find part b.ii in this mark scheme extract

Q2(c)(iii)	Any of: 1. Message (1) 2. startPos (Python) / pos (Java or C#) (1) 3. characters (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 9 of 13 · find part b.ii in this mark scheme extract

Q2(c)(iv)	newMessage (1)	Do not accept outMessage	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 10 of 13 · find part b.ii in this mark scheme extract

Q2(c)(v)	Response starts with 'c' (1) Response has six letters in sequence from 'educational' (1)	Marks can be awarded from Q2(c)(vi)	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 11 of 13 · find part b.ii in this mark scheme extract

Q2(c)(vi)	1. Returns substring / crops a string (1) 2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1)	Award two marks for: It produces a substring of the message starting at position startPos with length characters Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1) Accept a correct generic explanation of what a subprogram does for 1 mark	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 12 of 13 · find part b.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 2 of 23 · Question 2 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 13 of 13 · find part b.ii in this mark scheme extract

2(a)	Award 1 mark for each of: 1. set counter to 1 and guess to 0 (1) 2. set answer to random value between 1 and 10 (1) 3. request input of and accept an integer value for guess (1) 4. create a while loop with a correct condition (1) 5. increment the counter (1) 6. use if else selection to determine and display appropriate output message for incorrect guesses (1) 7. request input of and accept another integer value for a guess inside the loop (1) 8. display correct guess and count of guesses (1) 9. compiling without syntax errors (1) 10. coding meets all requirements of question (1)	Candidates are required to open the file Q02a in the code editor. Amended code should be saved as Q02aFINISHED. Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements.	(10)

Question	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 1 of 13 · find part c.i in this mark scheme extract

2(b)(i)	24.0 (1)	Accept 24	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 2 of 13 · find part c.i in this mark scheme extract

2(b)(ii)	One of: By the rules of precedence multiplication is done before addition. (1) Without the brackets the addition would be done to the total of the multiplication. (1) By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1) newSpeed and intialSpeed have to be added together before multiplying (by time). (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 3 of 13 · find part c.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 4 of 13 · find part c.i in this mark scheme extract

2(b)(iii)	Real (1) Because real x integer = real (1)	Accept double / float Accept 'because it could be a number with decimal places'	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 5 of 13 · find part c.i in this mark scheme extract

2(b)(iv)	27.375 (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 6 of 13 · find part c.i in this mark scheme extract

Q2(c)(i)	C#: 16 or 20 (1) Java: 16 or 21 (1) Python: 8 or 10 (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 7 of 13 · find part c.i in this mark scheme extract

Q2(c)(ii)	C#: 29 or 31 (1) Java: 24 or 26 (1) Python: 14 or 15 (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 8 of 13 · find part c.i in this mark scheme extract

Q2(c)(iii)	Any of: 1. Message (1) 2. startPos (Python) / pos (Java or C#) (1) 3. characters (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 9 of 13 · find part c.i in this mark scheme extract

Q2(c)(iv)	newMessage (1)	Do not accept outMessage	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 10 of 13 · find part c.i in this mark scheme extract

Q2(c)(v)	Response starts with 'c' (1) Response has six letters in sequence from 'educational' (1)	Marks can be awarded from Q2(c)(vi)	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 11 of 13 · find part c.i in this mark scheme extract

Q2(c)(vi)	1. Returns substring / crops a string (1) 2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1)	Award two marks for: It produces a substring of the message starting at position startPos with length characters Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1) Accept a correct generic explanation of what a subprogram does for 1 mark	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 12 of 13 · find part c.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 3 of 23 · Question 2 part c.i · 1 mark

2019 JUN · Paper 02 · extract 13 of 13 · find part c.i in this mark scheme extract

2(a)	Award 1 mark for each of: 1. set counter to 1 and guess to 0 (1) 2. set answer to random value between 1 and 10 (1) 3. request input of and accept an integer value for guess (1) 4. create a while loop with a correct condition (1) 5. increment the counter (1) 6. use if else selection to determine and display appropriate output message for incorrect guesses (1) 7. request input of and accept another integer value for a guess inside the loop (1) 8. display correct guess and count of guesses (1) 9. compiling without syntax errors (1) 10. coding meets all requirements of question (1)	Candidates are required to open the file Q02a in the code editor. Amended code should be saved as Q02aFINISHED. Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements.	(10)

Question	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 1 of 13 · find part c.ii in this mark scheme extract

2(b)(i)	24.0 (1)	Accept 24	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 2 of 13 · find part c.ii in this mark scheme extract

2(b)(ii)	One of: By the rules of precedence multiplication is done before addition. (1) Without the brackets the addition would be done to the total of the multiplication. (1) By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1) newSpeed and intialSpeed have to be added together before multiplying (by time). (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 3 of 13 · find part c.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 4 of 13 · find part c.ii in this mark scheme extract

2(b)(iii)	Real (1) Because real x integer = real (1)	Accept double / float Accept 'because it could be a number with decimal places'	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 5 of 13 · find part c.ii in this mark scheme extract

2(b)(iv)	27.375 (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 6 of 13 · find part c.ii in this mark scheme extract

Q2(c)(i)	C#: 16 or 20 (1) Java: 16 or 21 (1) Python: 8 or 10 (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 7 of 13 · find part c.ii in this mark scheme extract

Q2(c)(ii)	C#: 29 or 31 (1) Java: 24 or 26 (1) Python: 14 or 15 (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 8 of 13 · find part c.ii in this mark scheme extract

Q2(c)(iii)	Any of: 1. Message (1) 2. startPos (Python) / pos (Java or C#) (1) 3. characters (1)		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 9 of 13 · find part c.ii in this mark scheme extract

Q2(c)(iv)	newMessage (1)	Do not accept outMessage	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 10 of 13 · find part c.ii in this mark scheme extract

Q2(c)(v)	Response starts with 'c' (1) Response has six letters in sequence from 'educational' (1)	Marks can be awarded from Q2(c)(vi)	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 11 of 13 · find part c.ii in this mark scheme extract

Q2(c)(vi)	1. Returns substring / crops a string (1) 2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1)	Award two marks for: It produces a substring of the message starting at position startPos with length characters Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1) Accept a correct generic explanation of what a subprogram does for 1 mark	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 12 of 13 · find part c.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 4 of 23 · Question 2 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 13 of 13 · find part c.ii in this mark scheme extract

2(a)	Award 1 mark for each of: 1. set counter to 1 and guess to 0 (1) 2. set answer to random value between 1 and 10 (1) 3. request input of and accept an integer value for guess (1) 4. create a while loop with a correct condition (1) 5. increment the counter (1) 6. use if else selection to determine and display appropriate output message for incorrect guesses (1) 7. request input of and accept another integer value for a guess inside the loop (1) 8. display correct guess and count of guesses (1) 9. compiling without syntax errors (1) 10. coding meets all requirements of question (1)	Candidates are required to open the file Q02a in the code editor. Amended code should be saved as Q02aFINISHED. Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements.	(10)

Question	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 5 of 23 · Question 1 part a · 1 mark

2020 NOV · Paper 02 · extract 1 of 10 · find part a in this mark scheme extract

1 (a)	A1	**C** (equals) (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 5 of 23 · Question 1 part a · 1 mark

2020 NOV · Paper 02 · extract 2 of 10 · find part a in this mark scheme extract

1 (b)	B1	Award one mark for any of: • TRUE **and** FALSE (1) • Yes **and** No (1) • T **and** F (1) • 0 **and** 1 (1)	Ignore capitalisation	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 5 of 23 · Question 1 part a · 1 mark

2020 NOV · Paper 02 · extract 3 of 10 · find part a in this mark scheme extract

1 (d)	D1	Award one mark for any of: • (Symbolic name associated with) a value that may be changed (1) • A container used to store data (1) • A data store whose contents can change whilst a program is executing (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 5 of 23 · Question 1 part a · 1 mark

2020 NOV · Paper 02 · extract 4 of 10 · find part a in this mark scheme extract

1 (e) (i)	E1	Python 6 C# 12 Java 8		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 5 of 23 · Question 1 part a · 1 mark

2020 NOV · Paper 02 · extract 5 of 10 · find part a in this mark scheme extract

1 (e) (ii)	E2	Python 3 C# 7 Java 8		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 5 of 23 · Question 1 part a · 1 mark

2020 NOV · Paper 02 · extract 6 of 10 · find part a in this mark scheme extract

1 (e) (iii)	E3	Award one mark for any of: • amount (1) • x (1) • total (1) • count (1)	Accept listNumbers or numArray	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 5 of 23 · Question 1 part a · 1 mark

2020 NOV · Paper 02 · extract 7 of 10 · find part a in this mark scheme extract

1 (e) (iv)	E4	message		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 5 of 23 · Question 1 part a · 1 mark

2020 NOV · Paper 02 · extract 8 of 10 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 5 of 23 · Question 1 part a · 1 mark

2020 NOV · Paper 02 · extract 9 of 10 · find part a in this mark scheme extract

1 (c)	C1	2.1.5	Delete space in variable name (1)		(3)
	C2	2.1.5	Capitalise O in variable name (1)		
	C3	2.1.5	Correction to print statement (1)		
Code example					
Python					

```
numOne = 25
numTwo = 36
numThree = numOne * numTwo
print (numThree)
```

9 of 16

Type MS · Spot constructs, operators, keywords and comments

Answer 5 of 23 · Question 1 part a · 1 mark

2020 NOV · Paper 02 · extract 10 of 10 · find part a in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 6 of 23 · Question 1 part e.i · 1 mark

2020 NOV · Paper 02 · extract 1 of 10 · find part e.i in this mark scheme extract

1 (a)	A1	**C** (equals) (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 6 of 23 · Question 1 part e.i · 1 mark

2020 NOV · Paper 02 · extract 2 of 10 · find part e.i in this mark scheme extract

1 (b)	B1	Award one mark for any of: • TRUE **and** FALSE (1) • Yes **and** No (1) • T **and** F (1) • 0 **and** 1 (1)	Ignore capitalisation	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 6 of 23 · Question 1 part e.i · 1 mark

2020 NOV · Paper 02 · extract 3 of 10 · find part e.i in this mark scheme extract

1 (d)	D1	Award one mark for any of: • (Symbolic name associated with) a value that may be changed (1) • A container used to store data (1) • A data store whose contents can change whilst a program is executing (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 6 of 23 · Question 1 part e.i · 1 mark

2020 NOV · Paper 02 · extract 4 of 10 · find part e.i in this mark scheme extract

1 (e) (i)	E1	Python 6 C# 12 Java 8		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 6 of 23 · Question 1 part e.i · 1 mark

2020 NOV · Paper 02 · extract 5 of 10 · find part e.i in this mark scheme extract

1 (e) (ii)	E2	Python 3 C# 7 Java 3		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 6 of 23 · Question 1 part e.i · 1 mark

2020 NOV · Paper 02 · extract 6 of 10 · find part e.i in this mark scheme extract

1 (e) (iii)	E3	Award one mark for any of: • amount (1) • x (1) • total (1) • count (1)	Accept listNumbers or numArray	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 6 of 23 · Question 1 part e.i · 1 mark

2020 NOV · Paper 02 · extract 7 of 10 · find part e.i in this mark scheme extract

1 (e) (iv)	E4	message		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 6 of 23 · Question 1 part e.i · 1 mark

2020 NOV · Paper 02 · extract 8 of 10 · find part e.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 6 of 23 · Question 1 part e.i · 1 mark

2020 NOV · Paper 02 · extract 9 of 10 · find part e.i in this mark scheme extract

1 (c)	C1	2.1.5	Delete space in variable name (1)		(3)
	C2	2.1.5	Capitalise O in variable name (1)		
	C3	2.1.5	Correction to print statement (1)		
Code example					
Python					

```
1  numOne = 25
2  numTwo = 36
3  numThree = numOne * numTwo
4  print (numThree)
```

9 of 16

Type MS · Spot constructs, operators, keywords and comments

Answer 6 of 23 · Question 1 part e.i · 1 mark

2020 NOV · Paper 02 · extract 10 of 10 · find part e.i in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 7 of 23 · Question 1 part e.ii · 1 mark

2020 NOV · Paper 02 · extract 1 of 10 · find part e.ii in this mark scheme extract

1 (a)	A1	**C** (equals) (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 7 of 23 · Question 1 part e.ii · 1 mark

2020 NOV · Paper 02 · extract 2 of 10 · find part e.ii in this mark scheme extract

1 (b)	B1	Award one mark for any of: • TRUE **and** FALSE (1) • Yes **and** No (1) • T **and** F (1) • 0 **and** 1 (1)	Ignore capitalisation	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 7 of 23 · Question 1 part e.ii · 1 mark

2020 NOV · Paper 02 · extract 3 of 10 · find part e.ii in this mark scheme extract

1 (d)	D1	Award one mark for any of: • (Symbolic name associated with) a value that may be changed (1) • A container used to store data (1) • A data store whose contents can change whilst a program is executing (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 7 of 23 · Question 1 part e.ii · 1 mark

2020 NOV · Paper 02 · extract 4 of 10 · find part e.ii in this mark scheme extract

1 (e) (i)	E1	Python 6 C# 12 Java 8		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 7 of 23 · Question 1 part e.ii · 1 mark

2020 NOV · Paper 02 · extract 5 of 10 · find part e.ii in this mark scheme extract

1 (e) (ii)	E2	Python 3 C# 7 Java 3		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 7 of 23 · Question 1 part e.ii · 1 mark

2020 NOV · Paper 02 · extract 6 of 10 · find part e.ii in this mark scheme extract

1 (e) (iii)	E3	Award one mark for any of: • amount (1) • x (1) • total (1) • count (1)	Accept listNumbers or numArray	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 7 of 23 · Question 1 part e.ii · 1 mark

2020 NOV · Paper 02 · extract 7 of 10 · find part e.ii in this mark scheme extract

1 (e) (iv)	E4	message		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 7 of 23 · Question 1 part e.ii · 1 mark

2020 NOV · Paper 02 · extract 8 of 10 · find part e.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 7 of 23 · Question 1 part e.ii · 1 mark

2020 NOV · Paper 02 · extract 9 of 10 · find part e.ii in this mark scheme extract

1 (c)	C1	2.1.5	Delete space in variable name (1)		(3)
	C2	2.1.5	Capitalise O in variable name (1)		
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Code example					
Python					

```
numOne = 25
numTwo = 36
numThree = numOne * numTwo
print (numThree)
```

9 of 16

Type MS · Spot constructs, operators, keywords and comments

Answer 7 of 23 · Question 1 part e.ii · 1 mark

2020 NOV · Paper 02 · extract 10 of 10 · find part e.ii in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 8 of 23 · Question 1 part a · 1 mark

2021 JUN · Paper 02 · extract 1 of 10 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (a)	A1	**1. The only correct answer is B** ***A*** *is not correct because as it is an arithmetic operator* ***C*** *is not correct because as it is a relational operator* ***D*** *is not correct because as it is a relational operator*		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 8 of 23 · Question 1 part a · 1 mark

2021 JUN · Paper 02 · extract 2 of 10 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (b)	B1 B2	Award up to 2 marks for a linked description such as: • 1D represents items as a list (1), 2D as a table / 2D as rows and columns (1) • 1D is a row/column (1), 2D is a table / 2D has rows and columns (1) • Each element in 1D is a single value (1), each element in 2D is a 1D array (1) • 1D array can only store one type of element (1). 2D array can store multiple elements in it (1)	Accept an interpretation	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 8 of 23 · Question 1 part a · 1 mark

2021 JUN · Paper 02 · extract 3 of 10 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (c)	C1	Change num_twenties == to num_twenties = (1)		
	C2	The **left over** variable named the same in both places (1)		
	C3	Add + before str(left_over) over in final print statement (1)	Accept equivalent e.g. ,	(3)

Type MS · Spot constructs, operators, keywords and comments

Answer 8 of 23 · Question 1 part a · 1 mark

2021 JUN · Paper 02 · extract 4 of 10 · find part a in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
1 (d) (i)	D1	Award 1 mark for adding an appropriate comment at the end of the line where there is relational operator: `if vowel == letter:  # relational operator and selection`	May be on different line numbers The comment(s)	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 8 of 23 · Question 1 part a · 1 mark

2021 JUN · Paper 02 · extract 5 of 10 · find part a in this mark scheme extract

1 (d) (ii)	D2	Award one mark for adding an appropriate comment at the end of a line where iteration starts:
		`8 for letter in sentence:  # iteration starts`
		`9  for vowel in vowels: # iteration starts`
			added must clearly identify the component	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 8 of 23 · Question 1 part a · 1 mark

2021 JUN · Paper 02 · extract 6 of 10 · find part a in this mark scheme extract

1 (d) (iii)	D3	Award one mark for adding an appropriate comment at the end of the line where selection starts: `if vowel == letter:  # relational operator and selection`	

Type MS · Spot constructs, operators, keywords and comments

Answer 8 of 23 · Question 1 part a · 1 mark

2021 JUN · Paper 02 · extract 7 of 10 · find part a in this mark scheme extract

1 (d) (iv)	D4	Award one mark for adding an appropriate comment at the end of a line where a data structure is initialised:
		`vowels = ["a","e","i","o","u"] # data structure initalised`
		`numVowels = [0,0,0,0,0] # data structure initalised`
				(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 8 of 23 · Question 1 part a · 1 mark

2021 JUN · Paper 02 · extract 8 of 10 · find part a in this mark scheme extract

Type MS · Spot constructs, operators, keywords and comments

Answer 8 of 23 · Question 1 part a · 1 mark

2021 JUN · Paper 02 · extract 9 of 10 · find part a in this mark scheme extract

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<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td rowspan="6">2 (b)</td><td colspan="2">Award 1 mark for each correct condition.</td><td rowspan="5">Accept alternatives e.g. Line 11 booksSold <=4 etc.</td><td rowspan="5">

(4)</td></tr>
<tr><td></td><td>Condition</td><td>Output message</td></tr>
<tr><td>B1</td><td>Number of books sold is under 5 or profit made is under 5</td><td>Poor performance this week</td></tr>
<tr><td>B2</td><td>Number of books sold is over 20; profit made is at least 20</td><td>Sales and profit are excellent this week</td></tr>
<tr><td>B3</td><td>Number of books sold is at least 5; profit made is at least 10</td><td>Sales and profit are good this week</td></tr>
<tr><td>B4</td><td>All other inputs</td><td>Alert manager</td></tr>
<tr><td colspan="5">Code example</td></tr>
<tr><td>Python</td><td colspan="4">

```
11 if booksSold < 5 or profit < 5:
12     print("Poor performance this week")
13 elif booksSold > 20 and profit >= 20:
14     print("Sales and profit are excellent this week")
15 elif booksSold >=5 and profit >= 10:
16     print("Sales and profit are good this week")
17 else:
18     print("Alert manager")
```

</td></tr>
</table>

Type MS · Spot constructs, operators, keywords and comments

Answer 8 of 23 · Question 1 part a · 1 mark

2021 JUN · Paper 02 · extract 10 of 10 · find part a in this mark scheme extract

Type MS · Spot constructs, operators, keywords and comments

Answer 9 of 23 · Question 1 part d.i · 1 mark

2021 JUN · Paper 02 · extract 1 of 10 · find part d.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (a)	A1	**1. The only correct answer is B** ***A*** *is not correct because as it is an arithmetic operator* ***C*** *is not correct because as it is a relational operator* ***D*** *is not correct because as it is a relational operator*		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 9 of 23 · Question 1 part d.i · 1 mark

2021 JUN · Paper 02 · extract 2 of 10 · find part d.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (b)	B1 B2	Award up to 2 marks for a linked description such as: • 1D represents items as a list (1), 2D as a table / 2D as rows and columns (1) • 1D is a row/column (1), 2D is a table / 2D has rows and columns (1) • Each element in 1D is a single value (1), each element in 2D is a 1D array (1) • 1D array can only store one type of element (1). 2D array can store multiple elements in it (1)	Accept an interpretation	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 9 of 23 · Question 1 part d.i · 1 mark

2021 JUN · Paper 02 · extract 3 of 10 · find part d.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (c)	C1	Change num_twenties == to num_twenties = (1)		
	C2	The **left over** variable named the same in both places (1)		
	C3	Add + before str(left_over) over in final print statement (1)	Accept equivalent e.g. ,	(3)

Type MS · Spot constructs, operators, keywords and comments

Answer 9 of 23 · Question 1 part d.i · 1 mark

2021 JUN · Paper 02 · extract 4 of 10 · find part d.i in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
1 (d) (i)	D1	Award 1 mark for adding an appropriate comment at the end of the line where there is relational operator: `if vowel == letter:  # relational operator and selection`	May be on different line numbers The comment(s)	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 9 of 23 · Question 1 part d.i · 1 mark

2021 JUN · Paper 02 · extract 5 of 10 · find part d.i in this mark scheme extract

1 (d) (ii)	D2	Award one mark for adding an appropriate comment at the end of a line where iteration starts:
		`8 for letter in sentence:  # iteration starts`
		`9  for vowel in vowels: # iteration starts`
			added must clearly identify the component	

Type MS · Spot constructs, operators, keywords and comments

Answer 9 of 23 · Question 1 part d.i · 1 mark

2021 JUN · Paper 02 · extract 6 of 10 · find part d.i in this mark scheme extract

1 (d) (iii)	D3	Award one mark for adding an appropriate comment at the end of the line where selection starts: `if vowel == letter:  # relational operator and selection`

Type MS · Spot constructs, operators, keywords and comments

Answer 9 of 23 · Question 1 part d.i · 1 mark

2021 JUN · Paper 02 · extract 7 of 10 · find part d.i in this mark scheme extract

1 (d) (iv)	D4	Award one mark for adding an appropriate comment at the end of a line where a data structure is initialised:
		`vowels = ["a","e","i","o","u"] # data structure initalised`
		`numVowels = [0,0,0,0,0] # data structure initalised`
				(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 9 of 23 · Question 1 part d.i · 1 mark

2021 JUN · Paper 02 · extract 8 of 10 · find part d.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 9 of 23 · Question 1 part d.i · 1 mark

2021 JUN · Paper 02 · extract 9 of 10 · find part d.i in this mark scheme extract

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(4)</td></tr>
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<tr><td>B4</td><td>All other inputs</td><td>Alert manager</td></tr>
<tr><td colspan="5">Code example</td></tr>
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```
11 if booksSold < 5 or profit < 5:
12     print("Poor performance this week")
13 elif booksSold > 20 and profit >= 20:
14     print("Sales and profit are excellent this week")
15 elif booksSold >=5 and profit >= 10:
16     print("Sales and profit are good this week")
17 else:
18     print("Alert manager")
```

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Type MS · Spot constructs, operators, keywords and comments

Answer 9 of 23 · Question 1 part d.i · 1 mark

2021 JUN · Paper 02 · extract 10 of 10 · find part d.i in this mark scheme extract

Question	ans	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 10 of 23 · Question 1 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 1 of 10 · find part d.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (a)	A1	**1. The only correct answer is B** ***A*** *is not correct because as it is an arithmetic operator* ***C*** *is not correct because as it is a relational operator* ***D*** *is not correct because as it is a relational operator*		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 10 of 23 · Question 1 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 2 of 10 · find part d.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (b)	B1 B2	Award up to 2 marks for a linked description such as: • 1D represents items as a list (1), 2D as a table / 2D as rows and columns (1) • 1D is a row/column (1), 2D is a table / 2D has rows and columns (1) • Each element in 1D is a single value (1), each element in 2D is a 1D array (1) • 1D array can only store one type of element (1). 2D array can store multiple elements in it (1)	Accept an interpretation	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 10 of 23 · Question 1 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 3 of 10 · find part d.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (c)	C1	Change num_twenties == to num_twenties = (1)		
	C2	The **left over** variable named the same in both places (1)		
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Type MS · Spot constructs, operators, keywords and comments

Answer 10 of 23 · Question 1 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 4 of 10 · find part d.ii in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
1 (d) (i)	D1	Award 1 mark for adding an appropriate comment at the end of the line where there is relational operator: `if vowel == letter:  # relational operator and selection`	May be on different line numbers The comment(s)	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 10 of 23 · Question 1 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 5 of 10 · find part d.ii in this mark scheme extract

1 (d) (ii)	D2	Award one mark for adding an appropriate comment at the end of a line where iteration starts:
		`8 for letter in sentence:  # iteration starts`
		`9  for vowel in vowels: # iteration starts`
			added must clearly identify the component	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 10 of 23 · Question 1 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 6 of 10 · find part d.ii in this mark scheme extract

1 (d) (iii)	D3	Award one mark for adding an appropriate comment at the end of the line where selection starts: `if vowel == letter:  # relational operator and selection`

Type MS · Spot constructs, operators, keywords and comments

Answer 10 of 23 · Question 1 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 7 of 10 · find part d.ii in this mark scheme extract

1 (d) (iv)	D4	Award one mark for adding an appropriate comment at the end of a line where a data structure is initialised:
		`vowels = ["a","e","i","o","u"] # data structure initalised`
		`numVowels = [0,0,0,0,0] # data structure initalised`
				(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 10 of 23 · Question 1 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 8 of 10 · find part d.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 10 of 23 · Question 1 part d.ii · 1 mark

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<tr><td>B2</td><td>Number of books sold is over 20; profit made is at least 20</td><td>Sales and profit are excellent this week</td></tr>
<tr><td>B3</td><td>Number of books sold is at least 5; profit made is at least 10</td><td>Sales and profit are good this week</td></tr>
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```
11 if booksSold < 5 or profit < 5:
12     print("Poor performance this week")
13 elif booksSold > 20 and profit >= 20:
14     print("Sales and profit are excellent this week")
15 elif booksSold >=5 and profit >= 10:
16     print("Sales and profit are good this week")
17 else:
18     print("Alert manager")
```

</td></tr>
</table>

Type MS · Spot constructs, operators, keywords and comments

Answer 10 of 23 · Question 1 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 10 of 10 · find part d.ii in this mark scheme extract

Type MS · Spot constructs, operators, keywords and comments

Answer 11 of 23 · Question 1 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 1 of 10 · find part d.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (a)	A1	**1. The only correct answer is B** ***A*** *is not correct because as it is an arithmetic operator* ***C*** *is not correct because as it is a relational operator* ***D*** *is not correct because as it is a relational operator*		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 11 of 23 · Question 1 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 2 of 10 · find part d.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (b)	B1 B2	Award up to 2 marks for a linked description such as: • 1D represents items as a list (1), 2D as a table / 2D as rows and columns (1) • 1D is a row/column (1), 2D is a table / 2D has rows and columns (1) • Each element in 1D is a single value (1), each element in 2D is a 1D array (1) • 1D array can only store one type of element (1). 2D array can store multiple elements in it (1)	Accept an interpretation	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 11 of 23 · Question 1 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 3 of 10 · find part d.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (c)	C1	Change num_twenties == to num_twenties = (1)		
	C2	The **left over** variable named the same in both places (1)		
	C3	Add + before str(left_over) over in final print statement (1)	Accept equivalent e.g. ,	(3)

Type MS · Spot constructs, operators, keywords and comments

Answer 11 of 23 · Question 1 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 4 of 10 · find part d.iii in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
1 (d) (i)	D1	Award 1 mark for adding an appropriate comment at the end of the line where there is relational operator: `if vowel == letter:  # relational operator and selection`	May be on different line numbers The comment(s)	(1)

Type MS · Spot constructs, operators, keywords and comments

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2021 JUN · Paper 02 · extract 5 of 10 · find part d.iii in this mark scheme extract

1 (d) (ii)	D2	Award one mark for adding an appropriate comment at the end of a line where iteration starts:
		`8 for letter in sentence:  # iteration starts`
		`9  for vowel in vowels: # iteration starts`
			added must clearly identify the component

Type MS · Spot constructs, operators, keywords and comments

Answer 11 of 23 · Question 1 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 6 of 10 · find part d.iii in this mark scheme extract

1 (d) (iii)	D3	Award one mark for adding an appropriate comment at the end of the line where selection starts: `if vowel == letter:  # relational operator and selection`	

Type MS · Spot constructs, operators, keywords and comments

Answer 11 of 23 · Question 1 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 7 of 10 · find part d.iii in this mark scheme extract

1 (d) (iv)	D4	Award one mark for adding an appropriate comment at the end of a line where a data structure is initialised:
		`vowels = ["a","e","i","o","u"] # data structure initalised`
		`numVowels = [0,0,0,0,0] # data structure initalised`
				(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 11 of 23 · Question 1 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 8 of 10 · find part d.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 11 of 23 · Question 1 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 9 of 10 · find part d.iii in this mark scheme extract

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11 if booksSold < 5 or profit < 5:
12     print("Poor performance this week")
13 elif booksSold > 20 and profit >= 20:
14     print("Sales and profit are excellent this week")
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17 else:
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</td></tr>
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Type MS · Spot constructs, operators, keywords and comments

Answer 11 of 23 · Question 1 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 10 of 10 · find part d.iii in this mark scheme extract

Question	...	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 12 of 23 · Question 1 part d.iv · 1 mark

2021 JUN · Paper 02 · extract 1 of 10 · find part d.iv in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (a)	A1	**1. The only correct answer is B** ***A*** *is not correct because as it is an arithmetic operator* ***C*** *is not correct because as it is a relational operator* ***D*** *is not correct because as it is a relational operator*		(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 12 of 23 · Question 1 part d.iv · 1 mark

2021 JUN · Paper 02 · extract 2 of 10 · find part d.iv in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (b)	B1 B2	Award up to 2 marks for a linked description such as: • 1D represents items as a list (1), 2D as a table / 2D as rows and columns (1) • 1D is a row/column (1), 2D is a table / 2D has rows and columns (1) • Each element in 1D is a single value (1), each element in 2D is a 1D array (1) • 1D array can only store one type of element (1). 2D array can store multiple elements in it (1)	Accept an interpretation	(2)

Type MS · Spot constructs, operators, keywords and comments

Answer 12 of 23 · Question 1 part d.iv · 1 mark

2021 JUN · Paper 02 · extract 3 of 10 · find part d.iv in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1 (c)	C1	Change num_twenties == to num_twenties = (1)		
	C2	The **left over** variable named the same in both places (1)		
	C3	Add + before str(left_over) over in final print statement (1)	Accept equivalent e.g. ,	(3)

Type MS · Spot constructs, operators, keywords and comments

Answer 12 of 23 · Question 1 part d.iv · 1 mark

2021 JUN · Paper 02 · extract 4 of 10 · find part d.iv in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
1 (d) (i)	D1	Award 1 mark for adding an appropriate comment at the end of the line where there is relational operator: `10 if vowel == letter: # relational operator and selection`	May be on different line numbers The comment(s)	(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 12 of 23 · Question 1 part d.iv · 1 mark

2021 JUN · Paper 02 · extract 5 of 10 · find part d.iv in this mark scheme extract

1 (d) (ii)	D2	Award one mark for adding an appropriate comment at the end of a line where iteration starts:
		`8 for letter in sentence:  # iteration starts`
		`9  for vowel in vowels: # iteration starts`
			added must clearly identify the component

Type MS · Spot constructs, operators, keywords and comments

Answer 12 of 23 · Question 1 part d.iv · 1 mark

2021 JUN · Paper 02 · extract 6 of 10 · find part d.iv in this mark scheme extract

1 (d) (iii)	D3	Award one mark for adding an appropriate comment at the end of the line where selection starts: `if vowel == letter:  # relational operator and selection`

Type MS · Spot constructs, operators, keywords and comments

Answer 12 of 23 · Question 1 part d.iv · 1 mark

2021 JUN · Paper 02 · extract 7 of 10 · find part d.iv in this mark scheme extract

1 (d) (iv)	D4	Award one mark for adding an appropriate comment at the end of a line where a data structure is initialised:
		`vowels = ["a","e","i","o","u"] # data structure initalised`
		`numVowels = [0,0,0,0,0] # data structure initalised`
				(1)

Type MS · Spot constructs, operators, keywords and comments

Answer 12 of 23 · Question 1 part d.iv · 1 mark

2021 JUN · Paper 02 · extract 8 of 10 · find part d.iv in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 12 of 23 · Question 1 part d.iv · 1 mark

2021 JUN · Paper 02 · extract 9 of 10 · find part d.iv in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td rowspan="6">2 (b)</td><td colspan="2">Award 1 mark for each correct condition.</td><td rowspan="5">Accept alternatives e.g. Line 11 booksSold <=4 etc.</td><td rowspan="5">

(4)</td></tr>
<tr><td></td><td>Condition</td><td>Output message</td></tr>
<tr><td>B1</td><td>Number of books sold is under 5 or profit made is under 5</td><td>Poor performance this week</td></tr>
<tr><td>B2</td><td>Number of books sold is over 20; profit made is at least 20</td><td>Sales and profit are excellent this week</td></tr>
<tr><td>B3</td><td>Number of books sold is at least 5; profit made is at least 10</td><td>Sales and profit are good this week</td></tr>
<tr><td>B4</td><td>All other inputs</td><td>Alert manager</td></tr>
<tr><td colspan="5">Code example</td></tr>
<tr><td>Python</td><td colspan="4">

```
11 if booksSold < 5 or profit < 5:
12     print("Poor performance this week")
13 elif booksSold > 20 and profit >= 20:
14     print("Sales and profit are excellent this week")
15 elif booksSold >=5 and profit >= 10:
16     print("Sales and profit are good this week")
17 else:
18     print("Alert manager")
```

</td></tr>
</table>

Type MS · Spot constructs, operators, keywords and comments

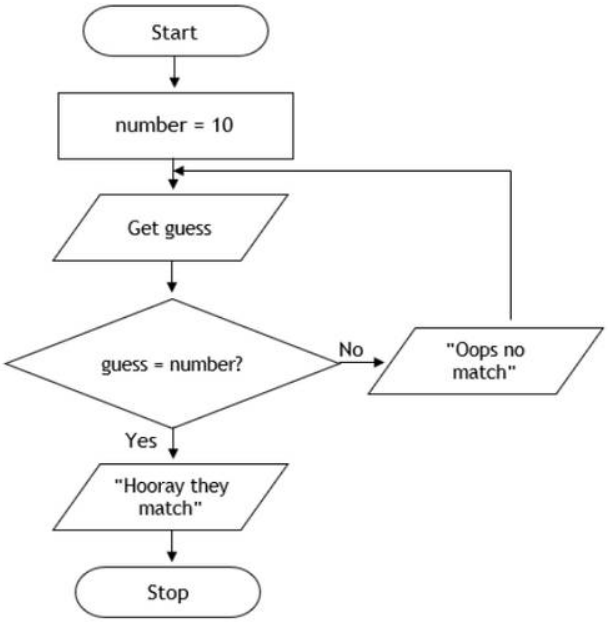
Answer 12 of 23 · Question 1 part d.iv · 1 mark

2021 JUN · Paper 02 · extract 10 of 10 · find part d.iv in this mark scheme extract

Type MS · Spot constructs, operators, keywords and comments

Answer 13 of 23 · Question 4 part b.i · 1 mark

2022 JUN · Paper 01 · extract 1 of 3 · find part b.i in this mark scheme extract

4(a)(i)	Award up to **five** marks for: • Start and stop terminators in the correct positions (1) 1-3 • Number set to 10 AND Get guess in the correct positions (1) 4-6 • Loop back to before Get guess and after number set to 10 if there is no match (1)Does not need to go via message box • Yes/no labels on decision match output messages (1) 4-6 • Correctly connected as in MS image, with at least 6 arrows correct (1) 7-9  	Boxes should be marked by content rather than shape.	5
4(a)(ii)	**The only correct answer is C** *A is not correct because a simulation is a completed program* *B is not correct because a cipher is a form of encryption* *D is not correct because a truth table is a method of testing an algorithm*		1

Type MS · Spot constructs, operators, keywords and comments

Answer 13 of 23 · Question 4 part b.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 3 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)(i)	Award **one** mark for: • D		1
4(b)(ii)	Award **one** mark for: • B		1

Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **four** marks for: • RedPoints column correct (1) • OrangePoints AND NumOranges columns correct (1) • Score correct in row 8 OR row 9 (1) -1 or below • Outputs correct and starting on the same row as the score OR the row below (1) can all be on one line		4
4(c)(ii)	Award **one** mark for: • 23		1

Colour	Score	Red Points	Orange Points	Num Oranges	Outputs
	0	0	0	0	
red		1			
orange			8	1	
red		2			
red		3			
orange			16	2	
-1					
	19				
					Score: 19
					Number of reds: 3
					Number of oranges: 16

Type MS · Spot constructs, operators, keywords and comments

Answer 13 of 23 · Question 4 part b.i · 1 mark

2022 JUN · Paper 01 · extract 3 of 3 · find part b.i in this mark scheme extract

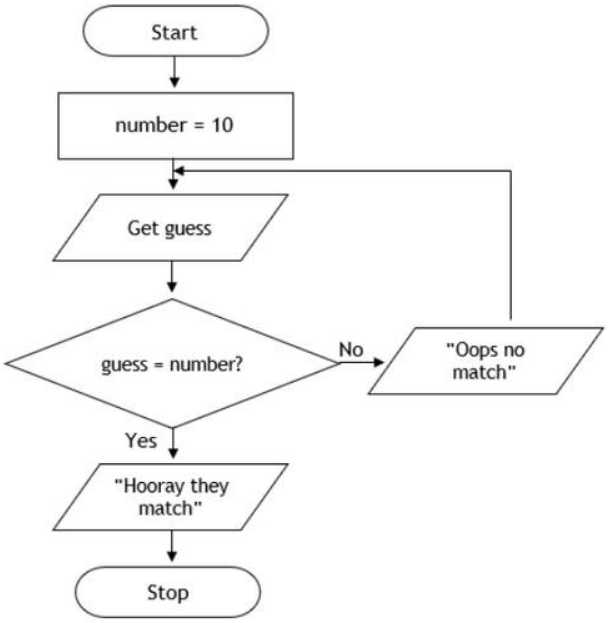
4(c)(iii)	Award **one** mark for: • Pseudododе that replaces OrangePoints with NumOranges on line 23 (1) SEND ("Number of oranges: "& NumOranges) TO DISPLAY		1
		Total for question 4	**14**

Question Number	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 14 of 23 · Question 4 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 3 · find part b.ii in this mark scheme extract

4(a)(i)	Award up to **five** marks for: • Start and stop terminators in the correct positions (1) 1-3 • Number set to 10 AND Get guess in the correct positions (1) 4-6 • Loop back to before Get guess and after number set to 10 if there is no match (1)Does not need to go via message box • Yes/no labels on decision match output messages (1) 4-6 • Correctly connected as in MS image, with at least 6 arrows correct (1) 7-9  	Boxes should be marked by content rather than shape.	5
4(a)(ii)	**The only correct answer is C** *A is not correct because a simulation is a completed program* *B is not correct because a cipher is a form of encryption* *D is not correct because a truth table is a method of testing an algorithm*		1

Type MS · Spot constructs, operators, keywords and comments

Answer 14 of 23 · Question 4 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 3 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)(i)	Award **one** mark for: • D		1
4(b)(ii)	Award **one** mark for: • B		1

Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **four** marks for: • RedPoints column correct (1) • OrangePoints AND NumOranges columns correct (1) • Score correct in row 8 OR row 9 (1) -1 or below • Outputs correct and starting on the same row as the score OR the row below (1) can all be on one line		4
4(c)(ii)	Award **one** mark for: • 23		1

Colour	Score	Red Points	Orange Points	Num Oranges	Outputs
	0	0	0	0	
red		1			
orange			8	1	
red		2			
red		3			
orange			16	2	
-1					
	19				
					Score: 19
					Number of reds: 3
					Number of oranges: 16

Type MS · Spot constructs, operators, keywords and comments

Answer 14 of 23 · Question 4 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 3 · find part b.ii in this mark scheme extract

4(c)(iii)	Award **one** mark for: • Pseudododе that replaces OrangePoints with NumOranges on line 23 (1) SEND ("Number of oranges: "& NumOranges) TO DISPLAY		1
		Total for question 4	**14**

Question Number	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 15 of 23 · Question 2 part a.i · 1 mark

2022 JUN · Paper 02 · extract 1 of 7 · find part a.i in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 15 of 23 · Question 2 part a.i · 1 mark

2022 JUN · Paper 02 · extract 2 of 7 · find part a.i in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 15 of 23 · Question 2 part a.i · 1 mark

2022 JUN · Paper 02 · extract 3 of 7 · find part a.i in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 15 of 23 · Question 2 part a.i · 1 mark

2022 JUN · Paper 02 · extract 4 of 7 · find part a.i in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 15 of 23 · Question 2 part a.i · 1 mark

2022 JUN · Paper 02 · extract 5 of 7 · find part a.i in this mark scheme extract

2 (a)(v)	A5	pNumber		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 15 of 23 · Question 2 part a.i · 1 mark

2022 JUN · Paper 02 · extract 6 of 7 · find part a.i in this mark scheme extract

2 (b)	**Award 1 mark for each of:**		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the input message.	(11)
	Evidence should be found in the Function			
	B1	Function interface is correct (1)		
	B2	Condition correct (for the number 1) (1)		
	B3	Loop initialised correctly (from 2 to the input number) (1)		
	B4	If condition correct (modulus) (1)		
	B5	Check returned (1)		
	Evidence should be found in the Main program			
	B6	Input of number stored as an integer variable (1)		
	B7	Input includes a space/colon / new line before the input value		
	B8	Function called correctly (1)		
	B9	Display includes number and message for either a prime number **OR** not (1)		
	Overall			
	B10	Compiling without syntax errors (1)		
	B11	Executing and producing the correct output (1)		
Code examples				
Python				

```python
# Write the function here
def checkPrime(pNumber):
    if pNumber == 1:
        check = False
    else:
        check = True
        for count in range(2,pNumber):
            if pNumber % count == 0 :
                check = False
    return check

# Write the main program here
number = int(input("Enter a number: "))
result = checkPrime(number)
if result == True:
    print(number,"is a prime number")
else:
    print(number,"is not a prime number")
```

Type MS · Spot constructs, operators, keywords and comments

Answer 15 of 23 · Question 2 part a.i · 1 mark

2022 JUN · Paper 02 · extract 7 of 7 · find part a.i in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 16 of 23 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 02 · extract 1 of 7 · find part a.ii in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 16 of 23 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 02 · extract 2 of 7 · find part a.ii in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 16 of 23 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 02 · extract 3 of 7 · find part a.ii in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 16 of 23 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 02 · extract 4 of 7 · find part a.ii in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 16 of 23 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 02 · extract 5 of 7 · find part a.ii in this mark scheme extract

2 (a)(v)	A5	pNumber		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 16 of 23 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 02 · extract 6 of 7 · find part a.ii in this mark scheme extract

2 (b)	**Award 1 mark for each of:**		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the input message.	(11)
	Evidence should be found in the Function			
	B1	Function interface is correct (1)		
	B2	Condition correct (for the number 1) (1)		
	B3	Loop initialised correctly (from 2 to the input number) (1)		
	B4	If condition correct (modulus) (1)		
	B5	Check returned (1)		
	Evidence should be found in the Main program			
	B6	Input of number stored as an integer variable (1)		
	B7	Input includes a space/colon / new line before the input value		
	B8	Function called correctly (1)		
	B9	Display includes number and message for either a prime number **OR** not (1)		
	Overall			
	B10	Compiling without syntax errors (1)		
	B11	Executing and producing the correct output (1)		
Code examples				
Python				

```python
# Write the function here
def checkPrime(pNumber):
    if pNumber == 1:
        check = False
    else:
        check = True
        for count in range(2,pNumber):
            if pNumber % count == 0 :
                check = False
    return check

# Write the main program here
number = int(input("Enter a number: "))
result = checkPrime(number)
if result == True:
    print(number,"is a prime number")
else:
    print(number,"is not a prime number")
```

Type MS · Spot constructs, operators, keywords and comments

Answer 16 of 23 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 02 · extract 7 of 7 · find part a.ii in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 17 of 23 · Question 2 part a.iii · 1 mark

2022 JUN · Paper 02 · extract 1 of 7 · find part a.iii in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 17 of 23 · Question 2 part a.iii · 1 mark

2022 JUN · Paper 02 · extract 2 of 7 · find part a.iii in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 17 of 23 · Question 2 part a.iii · 1 mark

2022 JUN · Paper 02 · extract 3 of 7 · find part a.iii in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 17 of 23 · Question 2 part a.iii · 1 mark

2022 JUN · Paper 02 · extract 4 of 7 · find part a.iii in this mark scheme extract



Type MS · Spot constructs, operators, keywords and comments

Answer 17 of 23 · Question 2 part a.iii · 1 mark

2022 JUN · Paper 02 · extract 5 of 7 · find part a.iii in this mark scheme extract

2 (a)(v)	A5	pNumber		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 17 of 23 · Question 2 part a.iii · 1 mark

2022 JUN · Paper 02 · extract 6 of 7 · find part a.iii in this mark scheme extract

2 (b)	**Award 1 mark for each of:**		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the input message.	(11)
	Evidence should be found in the Function			
	B1	Function interface is correct (1)		
	B2	Condition correct (for the number 1) (1)		
	B3	Loop initialised correctly (from 2 to the input number) (1)		
	B4	If condition correct (modulus) (1)		
	B5	Check returned (1)		
	Evidence should be found in the Main program			
	B6	Input of number stored as an integer variable (1)		
	B7	Input includes a space/colon / new line before the input value		
	B8	Function called correctly (1)		
	B9	Display includes number and message for either a prime number **OR** not (1)		
	Overall			
	B10	Compiling without syntax errors (1)		
	B11	Executing and producing the correct output (1)		
Code examples				
Python				

```python
# Write the function here
def checkPrime(pNumber):
    if pNumber == 1:
        check = False
    else:
        check = True
        for count in range(2,pNumber):
            if pNumber % count == 0 :
                check = False
    return check

# Write the main program here
number = int(input("Enter a number: "))
result = checkPrime(number)
if result == True:
    print(number,"is a prime number")
else:
    print(number,"is not a prime number")
```

Type MS · Spot constructs, operators, keywords and comments

Answer 17 of 23 · Question 2 part a.iii · 1 mark

2022 JUN · Paper 02 · extract 7 of 7 · find part a.iii in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 18 of 23 · Question 1 part b.i · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part b.i in this mark scheme extract

1(a)	A1 A2 A3	Award **one** mark for each correct cell: Input: Debit / Credit / Description Processing: Balance + Credit / Balance - Debit Output: Balance	Input: accept examples of inputs e.g. 10.23 (must be in Figure 1) Processing; mark first 3 things (ignore extra text) **if** correct e.g. **balance + credit** – debit (1) **balance + credit** + debit (1) balance – credit (0) accept examples from table e.g. 128.35 – 10.23 (1) Output: accept description of formula e.g. subtract debit from balance (1)	**3**

Question	mp	Answer	Additional Guidance	Mark
		Award **one** mark for each of:		
1(b)(i)	B1	This is 5%		**1**
1(b)(ii)	B2	or Accept || for C# and Java		**1**
1(b)(iii)	B3	grossAmt netAmt		**1**
1(b)(iv)	B4	TAX		**1**
1(b)(v)	B5	inGross		**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 18 of 23 · Question 1 part b.i · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part b.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1(c)(i)	C1	Award **one** mark for: To store a value that will not change (during program execution) (1)		**1**
1(c)(ii)	C2	Award **one** mark for any of the following: • It is less likely to be changed by accident (translation will error if attempt assignment)(1) • The value only needs to be changed in one place (for maintenance) (1) • It is easier to read/understand **the logic** of (1) • Ensure consistency of value if it has a name rather than the hard-coded same value repeatedly (typo) (1) • Can make code more efficient (optimisation / value stored rather than reference)	MP1 And MP4 are referring to programmer. Do **not** accept answers relating to user.	**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 18 of 23 · Question 1 part b.i · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part b.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 19 of 23 · Question 1 part b.ii · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part b.ii in this mark scheme extract

1(a)	A1 A2 A3	Award **one** mark for each correct cell: Input: Debit / Credit / Description Processing: Balance + Credit / Balance - Debit Output: Balance	Input: accept examples of inputs e.g. 10.23 (must be in Figure 1) Processing; mark first 3 things (ignore extra text) **if** correct e.g. **balance + credit** – debit (1) **balance + credit** + debit (1) balance – credit (0) accept examples from table e.g. 128.35 – 10.23 (1) Output: accept description of formula e.g. subtract debit from balance (1)	**3**

Question	mp	Answer	Additional Guidance	Mark
		Award **one** mark for each of:		
1(b)(i)	B1	This is 5%		**1**
1(b)(ii)	B2	or Accept || for C# and Java		**1**
1(b)(iii)	B3	grossAmt netAmt		**1**
1(b)(iv)	B4	TAX		**1**
1(b)(v)	B5	inGross		**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 19 of 23 · Question 1 part b.ii · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part b.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1(c)(i)	C1	Award **one** mark for: To store a value that will not change (during program execution) (1)		**1**
1(c)(ii)	C2	Award **one** mark for any of the following: • It is less likely to be changed by accident (translation will error if attempt assignment)(1) • The value only needs to be changed in one place (for maintenance) (1) • It is easier to read/understand **the logic** of (1) • Ensure consistency of value if it has a name rather than the hard-coded same value repeatedly (typo) (1) • Can make code more efficient (optimisation / value stored rather than reference)	MP1 And MP4 are referring to programmer. Do **not** accept answers relating to user.	**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 19 of 23 · Question 1 part b.ii · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part b.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 1 of 16 · find part c.i in this mark scheme extract

1(a)	A1	**The only correct answer is D** ***A*** *is not correct because it is the definition of decomposition* ***B*** *is not correct because it is a task undertaken during evaluation* ***C*** *is not correct because it describes a method for expressing algorithms*		**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 2 of 16 · find part c.i in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 3 of 16 · find part c.i in this mark scheme extract

1b		Award **one** mark for each of:		
	B1	Initialisation of myName to "" (1)	• `String myName = "";` • `String myName = "";` • `myName = ""` Allow null Allow single space between quotes " "	
	B2	Add a line to create myAge and initialise it to 0 (1)	• `int myAge = 0;` • `int myAge = 0;` • `myAge = 0` Allow any method	
	B3	Complete the test by inserting < (1)	• `if (myAge < 30)` • `if (myAge < 30)` • `if (myAge < 30):` Allow <= Can award reverse test only if welcome message contained within an 'else' part	
	B4	Add a line to display "Welcome" (1)	• `Console.WriteLine ("Welcome");` • `System.out.println ("Welcome");` • `print ("Welcome")` ignore indentation Allow any method	4

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 4 of 16 · find part c.i in this mark scheme extract

C#

```
// Q01bFINISHED

// Complete the line to initialise myName to an empty string
String myName = "";

// Add a line to create an integer variable called myAge and
//     initialise myAge to 0
int myAge = 0;

Console.WriteLine ("Enter your name: ");
myName = Console.ReadLine();
Console.WriteLine ("Enter your age: ");
myAge = Convert.ToInt32 (Console.ReadLine());

// Complete the test to check if the age is less than 30
if (myAge < 30)
{
    // Add a line to display a message that says "Welcome"
    Console.WriteLine ("Welcome");
}

```

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 5 of 16 · find part c.i in this mark scheme extract

Java

```java
// Q01bFINISHED

import java.util.Scanner;
public class Q01bFINISHED
{
    public static void main (String[] args)
    {
        Scanner myKeyboard = new Scanner (System.in);

        // Complete the line to initialise myName to an empty string
        String myName = "";

        // Add a line to create an integer variable called myAge and
        //     initialise myAge to 0
        int myAge = 0;

        System.out.println ("Enter your name: ");
        myName = myKeyboard.nextLine();
        System.out.println ("Enter your age: ");
        myAge = Integer.parseInt (myKeyboard.nextLine());

        // Complete the test to check if the age is less than 30
        if (myAge < 30)
        {
            // Add a line to display a message that says "Welcome"
            System.out.println ("Welcome");
        }
    }
}
```

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 6 of 16 · find part c.i in this mark scheme extract

Python

```
# Q01bFINISHED

# Complete the line to initialise myName to an empty string
myName = ""

# Add a line to create an integer variable called myAge and
#     initialise myAge to 0
myAge = 0

myName = input ("Enter your name: ")
myAge = int (input ("Enter your age: "))

# Complete the test to check if the age is less than 30
if (myAge < 30):

    # Add a line to display a message that says "Welcome"
    print ("Welcome")

```

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 7 of 16 · find part c.i in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 8 of 16 · find part c.i in this mark scheme extract

1c(i)		Award **one** mark for any of the following:		
	C1	C#: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Java: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Python: • # Q01c (1) • Q01c (1) • # Loop for 50 times (1) Loop for 50 times (1)	Ignore transcription errors	**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 9 of 16 · find part c.i in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 10 of 16 · find part c.i in this mark scheme extract

1c(ii)		Award **one** mark for any of the following:		
	C2	C#: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Java: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Python: • count = 0 (1) • total = 0 (1) • subtotal = 0 (1)	Ignore transcription errors	1
Question	**mp**	**Answer**	**Additional guidance**	**Mark**

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 11 of 16 · find part c.i in this mark scheme extract

1c(iii)		Award **one** mark for:		
	C3	• if (1)	Applies to all programming languages Ignore capitalisation	**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 12 of 16 · find part c.i in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 13 of 16 · find part c.i in this mark scheme extract

1c(iv)		Award **one** mark for:		
	C4	C#: • && (1) Java: • && (1) Python: • and (1)	Ignore capitalisation	1

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 14 of 16 · find part c.i in this mark scheme extract

1(d)	D1	**The only correct answer is B** ***A*** *is not correct because the data can be stored in an array as it is all the same data type (string)* ***C*** *is not correct because the data can be stored in an array as it is all the same data type (integer)* ***D*** *is not correct because the data can be stored in an array as it is all the same data type (real)*		**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 15 of 16 · find part c.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 20 of 23 · Question 1 part c.i · 1 mark

2024 JUN · Paper 02 · extract 16 of 16 · find part c.i in this mark scheme extract

<table>
<tr>
<td>1(e)</td>
<td>E1
E2</td>
<td>One mark for each correct cell:
<table>
<tr><th>Item</th><th>Data type</th></tr>
<tr><td>45.82</td><td>All programming languages:
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C#:
decimal/double
Java:
Double/double/BigDecimal
Python:
decimal</td></tr>
<tr><td>True</td><td>All programming languages:
Boolean/bool</td></tr>
</table></td>
<td>Ignore capitalisation
Do not accept string for True</td>
<td>2</td>
</tr>
</table>

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 1 of 16 · find part c.ii in this mark scheme extract

1(a)	A1	**The only correct answer is D** ***A*** *is not correct because it is the definition of decomposition* ***B*** *is not correct because it is a task undertaken during evaluation* ***C*** *is not correct because it describes a method for expressing algorithms*		**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 2 of 16 · find part c.ii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 3 of 16 · find part c.ii in this mark scheme extract

1b		Award **one** mark for each of:		
	B1	Initialisation of myName to "" (1)	• `String myName = "";` • `String myName = "";` • `myName = ""` Allow null Allow single space between quotes " "	
	B2	Add a line to create myAge and initialise it to 0 (1)	• `int myAge = 0;` • `int myAge = 0;` • `myAge = 0` Allow any method	
	B3	Complete the test by inserting < (1)	• `if (myAge < 30)` • `if (myAge < 30)` • `if (myAge < 30):` Allow <= Can award reverse test only if welcome message contained within an 'else' part	
	B4	Add a line to display "Welcome" (1)	• `Console.WriteLine ("Welcome");` • `System.out.println ("Welcome");` • `print ("Welcome")` ignore indentation Allow any method	4

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 4 of 16 · find part c.ii in this mark scheme extract

C#

```
// Q01bFINISHED

// Complete the line to initialise myName to an empty string
String myName = "";

// Add a line to create an integer variable called myAge and
//     initialise myAge to 0
int myAge = 0;

Console.WriteLine ("Enter your name: ");
myName = Console.ReadLine();
Console.WriteLine ("Enter your age: ");
myAge = Convert.ToInt32 (Console.ReadLine());

// Complete the test to check if the age is less than 30
if (myAge < 30)
{
    // Add a line to display a message that says "Welcome"
    Console.WriteLine ("Welcome");
}

```

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 5 of 16 · find part c.ii in this mark scheme extract

Java

```java
// Q01bFINISHED

import java.util.Scanner;
public class Q01bFINISHED
{
    public static void main (String[] args)
    {
        Scanner myKeyboard = new Scanner (System.in);

        // Complete the line to initialise myName to an empty string
        String myName = "";

        // Add a line to create an integer variable called myAge and
        //     initialise myAge to 0
        int myAge = 0;

        System.out.println ("Enter your name: ");
        myName = myKeyboard.nextLine();
        System.out.println ("Enter your age: ");
        myAge = Integer.parseInt (myKeyboard.nextLine());

        // Complete the test to check if the age is less than 30
        if (myAge < 30)
        {
            // Add a line to display a message that says "Welcome"
            System.out.println ("Welcome");
        }
    }
}
```

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 6 of 16 · find part c.ii in this mark scheme extract

Python

```python
# Q01bFINISHED

# Complete the line to initialise myName to an empty string
myName = ""

# Add a line to create an integer variable called myAge and
#     initialise myAge to 0
myAge = 0

myName = input ("Enter your name: ")
myAge = int (input ("Enter your age: "))

# Complete the test to check if the age is less than 30
if (myAge < 30):

    # Add a line to display a message that says "Welcome"
    print ("Welcome")

```

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 7 of 16 · find part c.ii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 8 of 16 · find part c.ii in this mark scheme extract

1c(i)		Award **one** mark for any of the following:		
	C1	C#: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Java: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Python: • # Q01c (1) • Q01c (1) • # Loop for 50 times (1) Loop for 50 times (1)	Ignore transcription errors	**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 9 of 16 · find part c.ii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 10 of 16 · find part c.ii in this mark scheme extract

1c(ii)		Award **one** mark for any of the following:		
	C2	C#: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Java: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Python: • count = 0 (1) • total = 0 (1) • subtotal = 0 (1)	Ignore transcription errors	1
Question	**mp**	**Answer**	**Additional guidance**	**Mark**

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 11 of 16 · find part c.ii in this mark scheme extract

1c(iii)		Award **one** mark for:		
	C3	• if (1)	Applies to all programming languages Ignore capitalisation	**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 12 of 16 · find part c.ii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 13 of 16 · find part c.ii in this mark scheme extract

1c(iv)		Award **one** mark for:		
	C4	C#: • && (1) Java: • && (1) Python: • and (1)	Ignore capitalisation	1

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 14 of 16 · find part c.ii in this mark scheme extract

1(d)	D1	**The only correct answer is B** ***A*** *is not correct because the data can be stored in an array as it is all the same data type (string)* ***C*** *is not correct because the data can be stored in an array as it is all the same data type (integer)* ***D*** *is not correct because the data can be stored in an array as it is all the same data type (real)*		**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 15 of 16 · find part c.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 21 of 23 · Question 1 part c.ii · 1 mark

2024 JUN · Paper 02 · extract 16 of 16 · find part c.ii in this mark scheme extract

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<td>2</td>
</tr>
</table>

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 1 of 16 · find part c.iii in this mark scheme extract

1(a)	A1	**The only correct answer is D** ***A*** *is not correct because it is the definition of decomposition* ***B*** *is not correct because it is a task undertaken during evaluation* ***C*** *is not correct because it describes a method for expressing algorithms*		**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 2 of 16 · find part c.iii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 3 of 16 · find part c.iii in this mark scheme extract

1b		Award **one** mark for each of:		
	B1	Initialisation of myName to "" (1)	• `String myName = "";` • `String myName = "";` • `myName = ""` Allow null Allow single space between quotes " "	
	B2	Add a line to create myAge and initialise it to 0 (1)	• `int myAge = 0;` • `int myAge = 0;` • `myAge = 0` Allow any method	
	B3	Complete the test by inserting < (1)	• `if (myAge < 30)` • `if (myAge < 30)` • `if (myAge < 30):` Allow <= Can award reverse test only if welcome message contained within an 'else' part	
	B4	Add a line to display "Welcome" (1)	• `Console.WriteLine ("Welcome");` • `System.out.println ("Welcome");` • `print ("Welcome")` ignore indentation Allow any method	4

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 4 of 16 · find part c.iii in this mark scheme extract

C#

```
// Q01bFINISHED

// Complete the line to initialise myName to an empty string
String myName = "";

// Add a line to create an integer variable called myAge and
//     initialise myAge to 0
int myAge = 0;

Console.WriteLine ("Enter your name: ");
myName = Console.ReadLine();
Console.WriteLine ("Enter your age: ");
myAge = Convert.ToInt32 (Console.ReadLine());

// Complete the test to check if the age is less than 30
if (myAge < 30)
{
    // Add a line to display a message that says "Welcome"
    Console.WriteLine ("Welcome");
}

```

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 5 of 16 · find part c.iii in this mark scheme extract

Java

```java
// Q01bFINISHED

import java.util.Scanner;
public class Q01bFINISHED
{
    public static void main (String[] args)
    {
        Scanner myKeyboard = new Scanner (System.in);

        // Complete the line to initialise myName to an empty string
        String myName = "";

        // Add a line to create an integer variable called myAge and
        //     initialise myAge to 0
        int myAge = 0;

        System.out.println ("Enter your name: ");
        myName = myKeyboard.nextLine();
        System.out.println ("Enter your age: ");
        myAge = Integer.parseInt (myKeyboard.nextLine());

        // Complete the test to check if the age is less than 30
        if (myAge < 30)
        {
            // Add a line to display a message that says "Welcome"
            System.out.println ("Welcome");
        }
    }
}
```

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 6 of 16 · find part c.iii in this mark scheme extract

Python

```python
# Q01bFINISHED

# Complete the line to initialise myName to an empty string
myName = ""

# Add a line to create an integer variable called myAge and
#     initialise myAge to 0
myAge = 0

myName = input ("Enter your name: ")
myAge = int (input ("Enter your age: "))

# Complete the test to check if the age is less than 30
if (myAge < 30):

    # Add a line to display a message that says "Welcome"
    print ("Welcome")

```

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 7 of 16 · find part c.iii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 8 of 16 · find part c.iii in this mark scheme extract

1c(i)		Award **one** mark for any of the following:		
	C1	C#: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Java: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Python: • # Q01c (1) • Q01c (1) • # Loop for 50 times (1) Loop for 50 times (1)	Ignore transcription errors	**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 9 of 16 · find part c.iii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 10 of 16 · find part c.iii in this mark scheme extract

1c(ii)		Award **one** mark for any of the following:		
	C2	C#: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Java: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Python: • count = 0 (1) • total = 0 (1) • subtotal = 0 (1)	Ignore transcription errors	1
Question	**mp**	**Answer**	**Additional guidance**	**Mark**

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 11 of 16 · find part c.iii in this mark scheme extract

1c(iii)		Award **one** mark for:		
	C3	• if (1)	Applies to all programming languages Ignore capitalisation	**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 12 of 16 · find part c.iii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 13 of 16 · find part c.iii in this mark scheme extract

1c(iv)		Award **one** mark for:		
	C4	C#: • && (1) Java: • && (1) Python: • and (1)	Ignore capitalisation	1

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 14 of 16 · find part c.iii in this mark scheme extract

1(d)	D1	**The only correct answer is B** ***A*** *is not correct because the data can be stored in an array as it is all the same data type (string)* ***C*** *is not correct because the data can be stored in an array as it is all the same data type (integer)* ***D*** *is not correct because the data can be stored in an array as it is all the same data type (real)*		**1**

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 15 of 16 · find part c.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 22 of 23 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 02 · extract 16 of 16 · find part c.iii in this mark scheme extract

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</tr>
</table>

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot constructs, operators, keywords and comments

Answer 23 of 23 · Question 1 part d.ii · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part d.ii in this mark scheme extract

1(a)	A1	**The only correct answer is B** ***A*** *is not correct because for is used in iteration* ***C*** *is not correct because new is used to declare an object* ***D*** *is not correct because while is used in repetition*		**1**

Question	mp	Answer	Additional Guidance	Mark
1(b)(i)	B1 B2 B3	Award one mark for each correct cell:		**3**

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.		✔	
The program does not translate.	✔		
The program crashes during execution.			✔

1 of 23

Type MS · Spot constructs, operators, keywords and comments

Answer 23 of 23 · Question 1 part d.ii · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part d.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(b)(ii)	B4	Line 15 - : added to end of line		
	B5	Line 20 – 0 changed to 1		
	B6	Line 22 – missing comma added between string and greenCount		3

```
# Q01bii

# initialise counters
ballCount = 0
greenCount = 0
redCount = 0

ballArray = ["green", "red", "green", "green", "red", "green","green", "green",
             "green", "green", "green", "red", "green", "red", "green", "red",
             "red", "green", "green", "red", "red", "green", "green", "red",
             "green", "red", "red", "green", "green", "red", "green", "green",
             "red", "green", "green", "green", "red", "green", "red","green"]

# iterate over ball array
for ball in ballArray:
    ballCount = ballCount + 1
    if ball == "green":
        greenCount = greenCount +1
    else:
        redCount = redCount + 1

print(" Green: ", greenCount, "balls \n", "Red: ", redCount,
      "balls \n", "Total: ", ballCount, "balls")
```

2 of 23

Type MS · Spot constructs, operators, keywords and comments

Answer 23 of 23 · Question 1 part d.ii · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part d.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(c)(i)	C1	Award one mark for: • To store a value that must not be changed/prevent accidental change of value • To simplify code maintenance/so that if the value must be updated the change only has to made in a single place • To avoid inconsistency error of specifying same value multiple times • To make code more readable/give meaningful names to values	Accidental change cannot refer to 'user'	1

Question	mp	Answer	Additional guidance	Mark
1(c)(ii)	C2 C3	Award **two** marks for a linked description such as: • A variable stores a single value (1) whereas an array stores multiple values (1)		2

Question	mp	Answer	Additional guidance	Mark
1(d)(i)	D1	Award **one** mark for: • getWages (1)	Accept 'main' Accept: def getWages def getWages (inSales, inAge) line 7 (Python)	1

3 of 23

Type MS · Spot constructs, operators, keywords and comments

Answer 23 of 23 · Question 1 part d.ii · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part d.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(d)(ii)	D2	Award **one** mark for: • and (1)	Do not accept line numbers Accept: &&	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iii)	D3	Award **one** mark for any of the following: • sales (1) • wages (1) • inSales (1)	Do not accept line numbers	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iv)	D4	Award **one** mark for any of the following: • inSales (1) • inAge (1)	Do not accept line numbers	1

4 of 23

Type MS · Spot constructs, operators, keywords and comments

Answer 23 of 23 · Question 1 part d.ii · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part d.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark